

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. (Second Year) Branch : Mechanical Semester : III Subject Code & Name: BTMEC302 Materials Science and Metallurgy Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		4
A)	Draw the crystal Structure of FCC and HCP.	CO1	2
B)	Define toughness and hardness.	CO2	2
C)	What is Gibbs phase rule?	CO3	2
Q.2	Solve Any Two of the following.		16
A)	Explain screw dislocation and edge dislocation in detail.	CO1	8
B)	What is importance of equilibrium diagram? Explain in detail.	CO3	8
C)	What are the different methods of hardness measurement? Explain any two.	CO2	8
Q. 3	Solve Any Two of the following.		16
A)	Draw the neat sketch Iron-iron carbide equilibrium diagram and explain phases existing on it.	CO3	8
B)	Explain hardening heat treatment process. Why tempering is only followed by hardening process?	CO4	8
C)	Classify surface hardening processes. Explain any two methods of hardening.	CO4	8
Q.4	Solve Any Two of the following.		16
A)	Write down short note on Sulphur printing test and Spark test.	CO5	8
B)	What are the types of Strengthening Mechanisms? Explain any one in detail.	CO6	8
C)	Write about the specimen preparation procedure in detail.	CO5	8
Q. 5	Solve Any Two of the following.		8
A)	Differentiate between annealing and normalizing processes.	CO3	4
B)	What are the steps involved in dye Penetrant inspection?	CO5	4
C)	What is objective of TTT diagram?	CO4	4
*** End ***			

The grid and the borders of the table will be hidden before final printing.